

When Everyone Has AI: A Theory of Marketing Advantage Under Algorithmic Parity

Algorithmic Parity, Differentiation Capital, and the Synthetic Content Paradox in the AI Era

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Abstract

Artificial intelligence (AI) is becoming shared marketing infrastructure rather than a proprietary capability. If competing firms use similar platforms, models and optimization systems, why should AI remain a source of advantage? This paper introduces Algorithmic Parity, the condition in which competitors have access to functionally similar AI for targeting, bidding, content generation, optimization and measurement, and argues that under parity the marginal advantage of generic AI adoption declines while the value of scarce complements rises. It defines five complements, jointly called Differentiation Capital: brand capital, proprietary data capital, human trust capital, causal learning capital and visibility capital. It also identifies a Synthetic Content Paradox: generative AI makes content production abundant while making meaningful attention scarcer. A stylized model yields eight falsifiable propositions. An audit of published secondary evidence (platform disclosures, adoption surveys, controlled productivity experiments, consumer surveys and search studies) finds the propositions consistent in direction but directly tested in few cases. The paper therefore specifies five pre-registered studies and two new metrics to test them. No new experimental data are reported in this version. The contribution is a theory, a measurement agenda and a managerial architecture for competing when everyone has AI.

Keywords: artificial intelligence; marketing strategy; competitive advantage; algorithmic parity; generative AI; advertising; brand equity; resource-based view; attention economy; marketing measurement

JEL classification: M31 (Marketing); M37 (Advertising); O33 (Technological Change: Choices and Consequences; Diffusion Processes); L21 (Business Objectives of the Firm); D83 (Search; Learning; Information and Knowledge)

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Status note: This version is a theory and research-design paper. Evidence cited is secondary and graded by reliability (Appendix C). The empirical studies in Section 7 are designed but not yet conducted.

1. The Puzzle

Between 2020 and 2025, the share of organizations using AI in at least one business function rose from about 50 percent to 88 percent, and the share using generative AI rose from 33 percent in 2023 to 79 percent in 2025 (McKinsey, 2024, 2025). On the advertising side, Meta reported that its end-to-end AI ad tools passed a USD 60 billion annual revenue run rate by late 2025, and that more than 4 million advertisers were using at least one of its generative AI tools by the end of 2024 (Marketing Dive, 2025; MediaCat, 2025). AI is no longer scarce.

This creates a puzzle for marketing strategy. In the strategy literature, a resource produces sustained advantage only if it is valuable, rare and hard to imitate or substitute (Barney, 1991). Access to AI for targeting, creative generation, optimization and analysis is valuable, but it is increasingly available to every competitor, often through the same platforms. If advantage cannot come from having the intelligence, where does it come from?

This question has a precedent. Carr (2003) argued that information technology becomes infrastructure as it spreads, and that infrastructure confers no lasting advantage. Later work showed that returns to IT depend on complementary organizational assets (Brynjolfsson & Hitt, 2003). Generative AI revives the debate in marketing with two new features. It commoditizes not only analysis and optimization but also content production, which marketers long treated as the creative, human part of the work. And much of the AI sits inside platforms that control the auction, the audience and the measurement.

1.1 Contribution

The paper makes five contributions.

- 1 It defines Algorithmic Parity and states when it should arise.
- 2 It specifies Differentiation Capital, five complements to AI that remain scarce under parity.
- 3 It identifies the Synthetic Content Paradox, which links content abundance to attention scarcity.
- 4 It derives eight falsifiable propositions from a stylized model and audits them against published evidence.
- 5 It specifies five pre-registered studies and two new metrics to test the theory, and sets out implications for firms, platforms and regulators.

1.2 What this paper is and is not

This is a theory and research-design paper. It builds on an evidence review of AI in advertising, but it reports no new experimental data. Where evidence is cited it is secondary, and much of it comes from vendor surveys and platform statements that are directional at best. The paper does not claim to have shown that the theory is correct. It claims that the theory is coherent, consistent with the direction of the available evidence, and testable.

2. Algorithmic Parity

2.1 Definition

Algorithmic Parity is the condition in which competing firms have access to functionally similar AI capabilities for targeting, bidding, content generation, optimization and measurement, so that differences in firm performance are not mainly explained by differences in access to AI.

The definition has three features. It concerns access to capability and not use or skill, since firms may differ in how well they use a shared tool. It is a property of a market and a time, and may hold for bidding long before it holds for creative work. And it is a matter of degree: parity is approached as adoption and tool convergence rise.

2.2 Mechanisms that produce parity

- **Diffusion.** Adoption follows an S-curve (Rogers, 2003). Generative AI scores high on trialability and low on complexity, which shortened the early-adopter phase.
- **Platform embedding.** Much advertising AI is delivered by a few platforms to all advertisers. Google's Performance Max, introduced in 2021, bundles targeting, bidding and creative selection, and was the most-used AI tool among Indian marketers in a 2025 survey (69 percent) (Dataslayer, 2026; Storyboard18, 2025). Meta's Advantage+ suite is available to every advertiser on the platform.
- **Model convergence.** Commercial models and image generators are available by subscription to any firm.
- **Delivery through incumbents.** Detection data suggest AI reaches most marketing teams as features inside platforms they already use, and not as standalone products (TechnologyChecker, 2026).

2.3 Evidence that parity is approaching

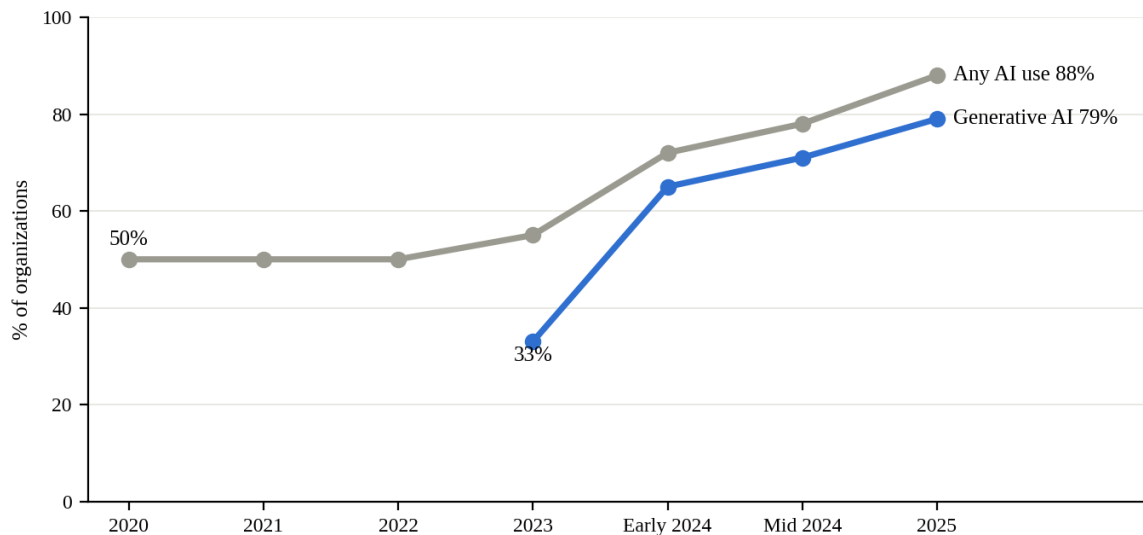


Figure 1. Organizational AI use rose from about 50% to 88% and generative AI from 33% to 79%. Source: McKinsey Global Surveys on the state of AI (2024, 2025). 2020 to 2022 are shown at the roughly 50% plateau McKinsey describes; its definition of AI use changed over time.

Adoption data are consistent with parity in basic use. Marketer-level surveys report that a large majority use AI: 86.4 percent of teams in HubSpot's 2026 survey and a recurring generative AI workflow for 87 percent of marketers in Salesforce data as summarized by Omnibound (TechnologyChecker, 2026; Omnibound, 2026). These measure use and not capability, and they come from vendor samples. They do not show that firms have the same capability, only that access is widespread. Depth remains uneven: only 7 percent of organizations report AI fully scaled (McKinsey, 2025), and 70 percent of CMOs say their processes are not mature enough to scale it (Gartner, 2026). Parity of access therefore coexists with large differences in use, and this gap is where Differentiation Capital operates.

2.4 Stylized path of advantage

Early on, adopters outperform non-adopters. As adoption spreads, the premium for adopting falls, and the sources of advantage move to complements. Figure 2 shows this as a stylized path and not as an empirical estimate.

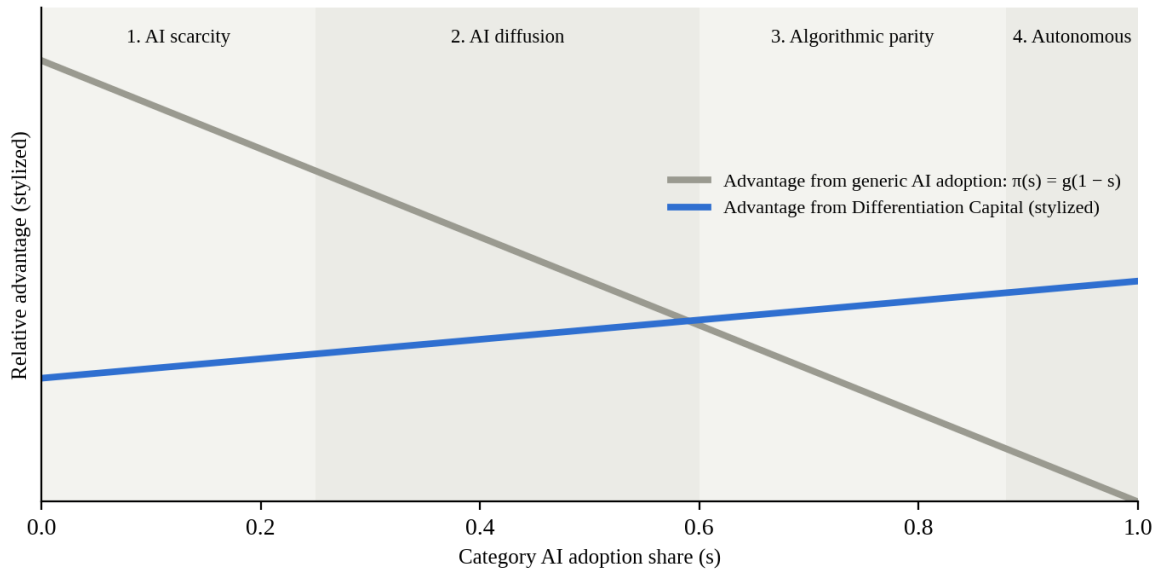


Figure 2. Stylized path of advantage as AI adoption in a category rises. Illustrative only; not estimated from data.

3. Differentiation Capital

3.1 Definition

Differentiation Capital is the set of resources that remain scarce when AI is widely and equally available, and that raise the return to AI when combined with it. It has five components.

Capital	Definition	Why it stays scarce	Observable proxies
Brand capital (B)	Distinctive assets, positioning, reputation, memory structures and creative identity	Built over years of consistent exposure; cannot be bought by licensing a model	Unaided awareness; recognition without logo; distinctive-asset scores
Proprietary data capital (D)	First-party customer data, purchase histories, proprietary insight and organizational knowledge	Owned by the firm; data-protection law limits transfer and reuse (for example the GDPR in the EU and the DPDP Act in India)	Share of customers with consented first-party records; data coverage and freshness
Human trust capital (H)	Human credibility, creators, employees, customer relationships and real-world evidence	Rests on relationships and proof that a model cannot generate	Trust scores; review volume and quality; human-presence ratings
Causal learning capital (C)	Experimentation, incrementality measurement and the ability to learn faster than rivals	Needs process, data access and discipline, not only tools	Share of spend under holdout; test velocity; gap between platform-reported and measured return
Visibility capital (V)	Discoverability in search, AI answers, recommendation systems and agent-mediated discovery	Depends on machine-readable authority and citable proprietary information	Citation share in AI answers; allowed crawlers; brand-search lift

The term "capital" is used in a broad sense. Brand capital and data capital resemble accumulated stocks. Causal learning capital and visibility capital are closer to capabilities or

accumulated positions, since they rest on routines (testing and measurement) and on earned authority in search and answer systems. The common feature is that each is a firm-specific stock or capability whose productive value persists across campaigns and raises the return to AI when combined with it. The paper keeps the single term for parsimony. Study 0 tests whether the five dimensions behave as one construct family or as distinct ones, and the label can be revised if they do not.

The theoretical lineage is direct. The five capitals are resource-based (Barney, 1991; Wernerfelt, 1984), brand equity (Keller, 1993; Sharp, 2010) and dynamic capabilities (Teece et al., 1997) ideas applied to the AI setting, with two additions specific to it: causal learning capital, because platform automation makes independent measurement rare, and visibility capital, because discovery is shifting toward generated answers.

3.2 The framework in one figure

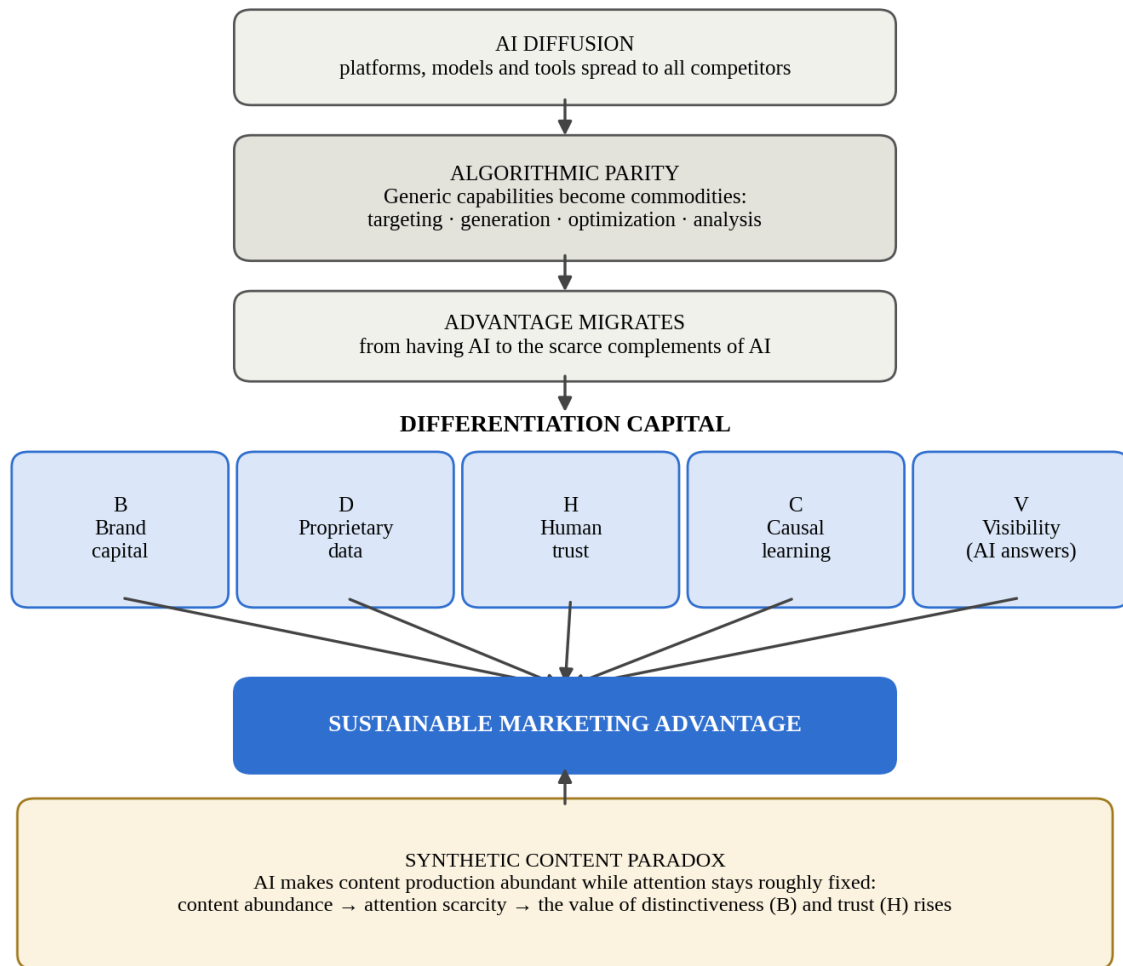


Figure 3. The Differentiation Capital framework: from AI diffusion to sustainable marketing advantage.

3.3 Relation to the earlier six differentiators

An earlier evidence review of AI in advertising (Nalla, 2026, Version 1) listed six differentiators: creative fundamentals, visible human craft, honest disclosure, proprietary data, disciplined testing and presence in AI answers. They map to the five capitals as follows: creative fundamentals to B; visible human craft and honest disclosure to H; proprietary data to D; disciplined testing to C; presence in AI answers to V. The five-capital version is intended to be more parsimonious and more general.

3.4 Why this is not simply the resource-based view applied to AI

A reviewer can fairly ask whether Algorithmic Parity and Differentiation Capital are new, or whether they relabel existing theory. The table states, for each related literature, what it already explains, what it leaves open in the AI setting, and what this paper adds. The claimed additions are conceptual and untested, and a reader may judge some of them incremental.

Existing theory	What it already explains	What it leaves open under AI	What this paper adds
Resource-based view (Barney, 1991; Wernerfelt, 1984)	Why valuable, rare, inimitable resources give lasting advantage	It does not say which marketing resources become rare when a powerful capability (AI) is rented by everyone from the same platforms	A specific claim about which complements stay scarce (B, D, H, C, V) and a condition (parity) under which the capability itself stops differentiating
IT commoditization (Carr, 2003)	Why technology that becomes universal loses strategic value	It treats the technology as a fixed infrastructure; generative AI also produces the content that carries the brand, and platforms control the auction and the measurement	The Synthetic Content Paradox, and platform dependency (P5) as a feature specific to advertising AI
Complementary assets (Teece, 1986; Brynjolfsson & Hitt, 2003)	Why returns to technology depend on complementary organizational assets	It does not identify the marketing-specific complements or predict how their value changes as content supply rises	A five-part complement set and propositions on how their value moves with adoption and with content abundance (P2, P7)
Dynamic capabilities (Teece et al., 1997)	Why firms that sense and reconfigure faster outperform	It does not address the measurement problem created by closed platform optimization	Causal learning capital (C) as a distinct capability, and P6 on measurement versus optimization skill
Brand equity and distinctive assets (Keller, 1993; Sharp, 2010)	Why strong memory structures and distinctive assets drive choice	It does not consider a market where rivals can produce unlimited similar content at near-zero cost	P3 and P7: creative quantity rises faster than distinctiveness, and distinctive assets gain value as supply rises
Attention economics (Simon, 1971)	Why information abundance creates attention scarcity	It does not link scarcity to the economics of AI-generated advertising content	Synthetic Content Paradox: value shifts from production to distinctiveness
Platform economics (Goldfarb & Tucker, 2019)	How digital platforms change advertising markets and measurement	It does not treat platform AI as a source of advertiser parity	P5, and visibility capital (V) as advantage in answer engines and agent-mediated discovery

The original elements are therefore: (1) the parity condition and the claim that advantage migrates from capability to complement in a market where the capability is rented from shared platforms; (2) the pairing of that claim with content abundance; (3) causal learning and visibility as named forms of capital; and (4) falsifiable propositions that distinguish the theory from its parents. If the studies in Section 7 fail to support P1, P3 and P7 in particular, the case for novelty weakens, and the theory should then be read as a useful synthesis.

4. The Synthetic Content Paradox

Generative AI lowers the cost of producing content. Klarna reported that AI cut its image development cycle from six weeks to seven days, produced more than 1,000 images in a quarter and handled about 80 percent of its copywriting (Marketing Dive, 2024). Coca-Cola's 2025 holiday advertisement reportedly used AI to generate more than 70,000 video clips in about 30 days (Retail Boss, 2025). Meta reports that advertisers using its video generation tools grew 20 percent in a single quarter, and the tools reached a USD 10 billion run rate (PPC Land, 2026).

Human attention does not scale with production. Simon (1971) observed that a wealth of information creates a poverty of attention. If content supply rises sharply and attention is roughly fixed, attention per unit of content falls.

Synthetic Content Paradox: the cheaper content becomes to produce, the harder meaningful attention becomes to obtain, so value shifts from production to distinctiveness.

4.1 Implications

- **Production is no longer the bottleneck.** Cost per asset falls for everyone. The scarce input becomes an idea or identity that is recognizably the brand's.
- **Quantity is not distinctiveness.** Kantar's analysis of Coca-Cola's AI advertisements concluded that effective creative principles remain the same whether or not AI is used (Research Live, 2025). More assets do not by themselves raise memorability.
- **Distinctive assets gain value.** As similar synthetic content fills feeds, a recognizable brand asset helps an ad to be identified and remembered (Sharp, 2010).
- **Trust becomes a filter.** Consumer comfort with brand AI fell from 57 to 46 percent between Q3 2023 and Q3 2024, and 59 percent named loss of the human touch as the main downside of brand AI (Qualtrics and Attest, via TechnologyChecker, 2026). Where content is abundant, signals of human authorship may carry more weight.

The paradox is a claim about direction. It does not say that AI content is ineffective. It says that when content abundance is common knowledge among competitors, the return to additional undifferentiated content falls.

5. A Stylized Model and Eight Propositions

5.1 Model

Let firm i 's marketing performance (for example, incremental revenue per unit of spend) be:

$$Y_i = A_i^\alpha \cdot K_i^\beta \cdot \varepsilon_i, \alpha, \beta > 0$$

where A_i is the AI capability available to firm i , K_i is its Differentiation Capital and ε_i is noise. Differentiation Capital is treated as a complementary composite, so that a missing component limits the whole:

$$K_i = B_i^{\theta_1} \cdot D_i^{\theta_2} \cdot H_i^{\theta_3} \cdot C_i^{\theta_4} \cdot V_i^{\theta_5}$$

Taking logs and expressing performance relative to the market average (subscript m):

$$\ln Y_i - \ln Y_m \approx \alpha (\ln A_i - \ln A_m) + \beta (\ln K_i - \ln K_m)$$

Under algorithmic parity, $A_i \approx \bar{A}$ for all firms, so the first term vanishes:

$$\ln Y_i - \ln Y_m \approx \beta (\ln K_i - \ln K_m)$$

In words: when everyone has similar intelligence, performance differences come from differences in Differentiation Capital. A simple heuristic form is $MA = A \times DC$ (marketing advantage as accessible intelligence times Differentiation Capital); under parity, differences in MA are driven by differences in DC.

Adoption dynamics can be written as an adopter premium that shrinks as the adoption share s rises:

$$\pi(s) = g \cdot (1 - s), 0 \leq s \leq 1$$

where g is the initial gain from adopting. As s approaches 1, π approaches 0. This is a deliberately simple form; the true shape is an empirical question (Study 1).

The model is a stylized device to organize predictions. It assumes constant elasticities and does not model competitive response, platform pricing or endogenous investment in K . Those are natural extensions.

5.2 Propositions

#	Proposition	Mechanism	Falsified if
P1	AI diffusion: the marginal performance advantage from generic AI adoption declines as category-level adoption rises	Imitation and convergence (π falls with s)	Adoption premium is flat or rising as category adoption rises
P2	Proprietary complement: the return to AI rises when it is combined with proprietary firm data and knowledge	Complementarity of A and K	No interaction between AI use and proprietary inputs
P3	Creative parity: generative AI increases creative quantity faster than creative distinctiveness	Shared models produce similar outputs	Distinctiveness rises in proportion to output
P4	Human salience: human involvement raises trust more when the interaction has high identity, emotional or credibility demands	Signals of human authorship matter most for such tasks	Human involvement has equal effect on low- and high-demand tasks
P5	Platform dependency: greater reliance on platform-controlled AI raises execution efficiency but lowers advertiser observability and strategic control	Closed optimization and platform-reported measurement	Observability does not fall as automation rises
P6	Measurement advantage: under parity, superior causal measurement creates more durable advantage than superior platform optimization	Learning compounds; optimization converges	Optimization skill predicts durable performance better than measurement skill
P7	Brand capital: the value of distinctive brand assets rises as generative content supply rises	Synthetic Content Paradox	Value of distinctive assets does not rise with supply
P8	AI visibility: as discovery shifts to generated answers and agents, visibility depends increasingly on machine-readable authority and citable proprietary information	Answer engines cite structured, authoritative sources	Citation in AI answers is unrelated to authority and structured proprietary content

6. Audit of Published Evidence

This section asks whether published secondary evidence is consistent with each proposition. It is not a test. Most sources are vendor surveys or platform statements, so a "consistent" rating means that the direction of the evidence is compatible with the proposition and that no contrary evidence was found, and not that the proposition is supported.

#	Evidence compatible with the proposition	Limits and contrary evidence	Status
P1	Adoption rose from about 50% to 88% (McKinsey, 2025); Meta's generative tools reached 4 million advertisers (MediaCat, 2025); the Meta price per ad rose 10% in Q3 2025 and 6% in Q4 2025, which is compatible with competition capturing gains (Marketing Dive, 2025; PPC Land, 2026)	No study measures the adopter premium against adoption share; price rises may reflect other causes	No direct test

#	Evidence compatible with the proposition	Limits and contrary evidence	Status
P2	Gartner: AI-mature organizations allocate 21.3% of budgets to AI against 15.3% on average and run larger budgets (Gartner, 2026); Brynjolfsson and Hitt (2003) show IT returns depend on complements	Correlational; mature firms may differ in other ways; not specific to proprietary data	Partly consistent
P3	Klarna: over 1,000 images in a quarter at lower cost (Marketing Dive, 2024); Kantar finds creative principles unchanged with AI (Research Live, 2025)	No study measures distinctiveness against output volume	No direct test
P4	Comfort with AI in ads is lowest for virtual brand ambassadors (34%) and higher for behind-the-scenes uses (43% for taglines) (YouGov, via TechnologyChecker, 2026); Luo et al. (2019) find chatbot disclosure reduced purchases in some conditions	Survey stated preferences; the field experiment is on chatbots	Partly consistent
P5	Practitioners describe lower visibility as automation improves (Dataslayer, 2026); Meta's incremental attribution feature reached a multi-billion-dollar run rate in seven months (PPC Land, 2026)	No measure of observability exists in the sources; platform statements	Partly consistent
P6	Only 41% of marketers can prove AI ROI (Konabayev, 2026); Klarna documented cost savings but not incremental revenue	Survey of one vendor's audience; no comparison of advantage by measurement skill	Partly consistent
P7	Kantar's testing suggests distinct, emotionally resonant creative drives response with or without AI (Research Live, 2025); Coca-Cola's public reaction (Campaign Asia, 2024)	No study varies content supply and measures value of distinctive assets	No direct test
P8	Search-purpose AI crawling rose from 1.32% to 8.26% (Cloudflare Radar, via TechnologyChecker, 2026); AI Overviews reduced click-through by about 58% (Ahrefs, 2026); Seer reports ChatGPT referrals convert at 15.9% against 1.76% for Google organic (Beancount, 2026)	Single-source conversion figure; Google disputes Pew's methodology; no study links authority or structured content to citation	Partly consistent

The audit has a clear message. The evidence is compatible with the direction of every proposition, and no source reviewed contradicts any of them. But for P1, P3 and P7, the three propositions that carry the core of the theory, no direct test exists. This is why Section 7 specifies original studies.

7. Empirical Program: Five Pre-Registered Studies

The five studies below are designed to test the propositions. They have not been conducted. Hypotheses, measures and analysis plans are to be registered on a public pre-registration platform before data collection, and results will be reported whether or not they support the theory.

7.1 Study 0: Develop and validate a Differentiation Capital Index

The five capitals are defined with proxies (Section 3.1) but have no validated scales. Study 0 follows the scale-development sequence of Churchill (1979).

- **Item generation.** Draft items for each capital from the literature and from 15 to 20 semi-structured interviews with senior marketers across categories and firm sizes, including India.
- **Content validity.** A panel of 10 to 15 academic and practitioner experts rates item relevance; items are retained using Lawshe's (1975) content validity ratio.
- **Exploratory factor analysis.** Pilot survey of about 300 marketing decision-makers to test whether items load on five factors (B, D, H, C, V).
- **Confirmatory factor analysis.** An independent sample of about 300 to test the five-factor model against one-factor and alternative models.
- **Reliability and validity.** Cronbach's alpha and McDonald's omega at or above 0.70; average variance extracted at or above 0.50 for convergent validity; discriminant validity using Fornell and Larcker (1981) and the heterotrait-monotrait ratio (Henseler et al., 2015), with ratios below 0.85.
- **Common method bias.** Procedural remedies and statistical checks (Podsakoff et al., 2003), including a marker variable.
- **Predictive validity.** Association of the index with firm performance outcomes, using objective measures where respondents can supply them and self-reported measures otherwise, controlling for firm size, category and AI use.
- **Index form.** Two forms will be compared: an additive index, $DCI = w_B B + w_D D + w_H H + w_C C + w_V V$, and the multiplicative composite of Section 5.1, which treats the capitals as complements. Whether the data favor one form is itself a test of the complementarity assumption.
- **Hypothesis.** H_{DC} : the five dimensions of Differentiation Capital are empirically distinct, and each provides incremental predictive validity beyond the others and beyond generic AI use.

7.2 Study 1: Does the AI adoption advantage decline with category adoption? (P1)

- **Question.** Does the performance premium from adopting AI tools fall as the share of competitors using them rises?
- **Design.** Panel of advertisers observed over multiple quarters, using staggered adoption of specific AI features. Event-study and difference-in-differences estimation, with the adopter premium interacted with category adoption share.

- **Data.** Advertiser-level performance data from partner platforms or agencies, or a repeated marketer survey with firm performance measures. A data-sharing agreement is the main practical dependency.
- **Hypothesis.** H1: the interaction between own adoption and category adoption is negative.
- **Outcome.** Return on ad spend or cost per incremental conversion; secondary outcomes include share of voice.

7.3 Study 2: Human, AI and Human+AI creative (P3, P4, P7)

- **Design.** Between-subjects online experiment. Identical creative briefs for at least four product categories are executed in three conditions: human only, AI only and Human+AI. A second factor varies disclosure of AI use (disclosed or not disclosed), and a third varies brand familiarity.
- **Sample.** About 3,000 participants in total, about 1,000 per condition. This gives about 80 percent power to detect an effect of $d = 0.15$ in a pairwise comparison at a Bonferroni-adjusted alpha of 0.0167; power for the full mixed model will be assessed by simulation before registration.
- **Measures.** Attention (eye tracking or attention-check proxy), recall, brand recognition with and without the logo, trust, purchase intention, emotion, perceived authenticity and perceived distinctiveness (independent coders and consumer ratings).
- **Hypotheses.** H3: AI-only creative has lower distinctiveness than human-only at similar quality ratings. H4: the human-involvement effect on trust is larger for high-identity briefs. H7: the advantage of distinctive assets increases when the comparison set contains more synthetic content.
- **Test of Human+AI.** Whether Human+AI exceeds Human only and exceeds AI only, and under which conditions.

7.4 Study 3: Do proprietary inputs improve AI advertising? (P2)

- **Design.** Randomized campaign experiment with partner advertisers. Within each advertiser, audiences or geographies are randomly assigned to (a) AI creative and targeting using generic inputs only, or (b) the same tools given the advertiser's brand knowledge base and first-party data signals. Holdouts provide the baseline.
- **Outcome.** Incremental conversions and incremental revenue per unit of spend; brand lift in a subsample.

- **Hypothesis.** H2: arm (b) outperforms arm (a), and the gap is larger for firms with richer first-party data.
- **Analysis.** Advertiser fixed effects with cluster-robust standard errors; heterogeneity by data richness and category.

7.5 Study 4: Does independent measurement change conclusions? (P5, P6)

- **Design.** For a sample of campaigns on automated platform products, compare platform-reported return on ad spend with incrementality estimates from holdout or geo-lift tests.
- **Outcome.** The gap between reported and incremental return, by campaign type and degree of automation.
- **Hypotheses.** H5: the gap is positive on average and larger for more automated campaign types. H6: advertisers with a higher share of spend under holdout show smaller variance in realized return over time.

7.6 Two proposed metrics

The theory suggests that cost per asset is the wrong productivity measure when content is abundant. Two alternatives are proposed.

$$DE = D_b / C_p \text{ (Distinctiveness Efficiency)}$$

where D_b is a measure of brand distinctiveness (for example, correct brand attribution without the logo) and C_p is creative production cost.

$$CP = \Delta R / C_{\text{total}} \text{ (Creative Productivity)}$$

where ΔR is the incremental brand or revenue effect from a creative set and C_{total} is its total cost, including review labor and media. These metrics let a firm compare, for example, 10,000 AI-generated advertisements with 100 memorable ones on outcomes and not on volume. They require measurement of incremental effect, which is the item most often missing in published claims.

8. Implications for Marketing Theory

- **Resource-based view.** The theory implies that the unit of advantage is not the technology but the complement. It extends Barney (1991) to a setting where the most powerful capability is rented from platforms.

- **Brand equity.** Keller (1993) and Sharp (2010) treat memory structures and distinctive assets as drivers of choice. The Synthetic Content Paradox predicts that their value rises when content is abundant.
- **Marketing capabilities and dynamic capabilities.** Causal learning capital recasts measurement as a capability that compounds, aligning with Teece et al. (1997).
- **Advertising and consumer psychology.** P3 and P4 give testable claims about distinctiveness and authenticity under disclosure.
- **Platform economics.** P5 treats platform AI as an input with a transparency cost and invites work on how gains are shared between platform and advertiser (Goldfarb & Tucker, 2019).
- **Information technology and strategy.** The theory is a marketing-specific test of the Carr (2003) versus complements debate (Brynjolfsson & Hitt, 2003).

9. Implications for Managers

If the theory is right, firms should invest in AI to reach parity and in Differentiation Capital to exceed it.

Capital	What to build	First actions
Brand (B)	A small set of distinctive, consistently used assets and a documented brand voice	Audit distinctive assets; encode brand rules for AI tools; keep flagship creative human-led
Data (D)	Consented first-party data and a knowledge base AI tools can use	Map data under DPDP and GDPR duties; feed platforms first-party signals; build an internal brand knowledge base
Human trust (H)	Visible human craft, real evidence and honest disclosure	Keep real people and proof where audiences expect them; design AI disclosure (EU Article 50 applies from 2 August 2026); review every published AI output
Causal learning (C)	Routine incrementality testing	Hold out audience or geography for major campaigns; reconcile platform-reported results to sales data quarterly; compare AI and human creative in the same test
Visibility (V)	Presence in AI answers	Allow search-purpose crawlers; publish structured, citable content and original research; track citations and brand-search lift

Gartner's 2026 data suggest that most firms are not yet organized to do this: 70 percent of CMOs say their processes are immature for scaling AI, 56 percent say they lack budget, and budgets stayed flat at 7.7 to 7.8 percent of revenue (Gartner, 2026). The architecture above therefore has to be funded by reallocation. Appendix A maps more than twenty marketing tactics to the capitals they build.

10. Implications for Platforms and Regulators

- **Platforms.** If P5 holds, advertisers will demand observability. Platforms could provide incrementality tools, independent measurement access and clearer reporting. Meta's incremental attribution feature, which reached a multi-billion-dollar run rate within seven months, suggests that demand exists (PPC Land, 2026).
- **Regulators.** The EU AI Act's Article 50 transparency duties have applied since 2 August 2026, covering chatbot disclosure, deepfake labelling and labelling of AI-generated text on matters of public interest unless a human takes editorial responsibility (Usercentrics, 2026; McCann FitzGerald, 2026). India's DPDP Rules, notified on 13 November 2025, phase in to 14 May 2027 and require clear, independently understandable notices (Legal500, 2025). The theory predicts that disclosure and data rules raise the value of D and H, because they restrict substitutes. These are legal facts current at the time of writing and should be rechecked.
- **Answer engines.** Evidence that AI answers reduce clicks by roughly 30 to 90 percent on affected queries across studies (Beancount, 2026; Ahrefs, 2026; Bosio Digital, 2026) raises questions about how value is shared between answer engines and the sources they draw on.

11. The Future: Agentic Competition

Agentic AI, which plans and executes multi-step tasks, is early in marketing. Salesforce's 2026 survey, as summarized by Usecarly (2026), found 13 percent of marketers using agentic AI, and HubSpot's 2026 data, as summarized by Voxbooster (2026), found 19.2 percent using agents end to end. Meta has said that each business will have an AI assistant that remembers its objectives, and it has announced agentic shopping tools (PPC Land, 2026).

If agents optimize campaigns continuously, parity in execution deepens, and human advantage moves further upstream: to strategy, identity, proprietary knowledge, governance and trust. Two scenarios follow. In an embedded-and-governed scenario, AI becomes infrastructure and advantage comes from Differentiation Capital. In a platform-capture scenario, platforms and answer engines capture most of the value and advertisers' share declines. A trust-correction scenario, in which consumer backlash and regulation slow customer-facing AI, would raise the value of H. The signposts to watch are the share of marketers who can prove ROI (41 percent in one 2026 survey), consumer comfort with brand AI (46 percent in 2024), the zero-click share (69 percent in one 2025 measurement) and AI Overview reach (48 percent of Google queries in April 2026) (Konabayev, 2026; TechnologyChecker, 2026; Beancount, 2026; DataRefs, 2026).

12. Limitations and Conclusion

12.1 Limitations

- **No original data.** Version 2 reports no experiments. The propositions are untested, and P1, P3 and P7 have no direct evidence.
- **Secondary and vendor sources.** Many statistics came through republishers; original reports were often not read. Platform figures are management statements.
- **Stylized model.** Constant elasticities, a composite capital index and no strategic interaction.
- **Construct measurement.** The capitals are defined with proxies, but validated scales do not yet exist. Study 0 is designed to address this.
- **Source quality.** Several consequential figures come from vendor surveys or republishing sites. Before any journal version, they should be replaced by original reports; this paper does not claim that has been done.
- **Alternative explanations.** Advantage under parity could also come from price, distribution, product quality and operations, which the model does not separate from marketing complements.
- **Boundary conditions.** Parity may not hold for firms without data, skills or budget; some AI capabilities may remain proprietary (for example, in-house models trained on unique data), which would blur the line between A and K.
- **Legal references.** Descriptions of the EU AI Act, the GDPR and India's DPDP regime are summaries for research context and have not been reviewed by legal counsel.
- **Time boundary.** Evidence runs to early October 2026, and regulation and platform products change quickly.

12.2 Conclusion

AI changes the economic location of marketing advantage. When execution, analysis and optimization are available to everyone, advantage moves to what AI does not supply: distinctive brands, proprietary data, human trust, the ability to learn causally and visibility to machines. At the same time, content abundance makes attention scarcer, which raises the value of being distinctive. The theory is simple enough to test and falsify, and this paper sets out how. If the studies in Section 7 support it, the practical message is that the question for marketers is no longer whether to adopt AI, but where advantage resides once everyone has it.

Appendix A. Marketing Tactic Map by Capital

The map lists tactics found in practice, the capital each mainly builds, the role of AI and the evidence found. Evidence strength: Moderate (platform, vendor or market data), Limited (anecdotal or headline only), None retrieved (author's judgment).

Tactic	Main capital	Role of AI	Evidence in this study
Positioning and distinctive brand assets	B	Low	Moderate: Kantar finding that creative fundamentals hold with or without AI
Brand storytelling and purpose	B, H	Low to medium	Moderate: comfort with brand AI fell to 46%
Original research and thought leadership	B, V	Medium	None retrieved on effect
Search (SEO) and generative engine optimization	V	High	Moderate: click losses of 30% to 90%; search-purpose crawling 8.26%
Short-form and long-form video	B	High	Moderate: digital video USD 78.0bn, up 25.4%; Reels run rate above USD 50bn
Creator and influencer partnerships	H	Medium	Moderate: US spend USD 37bn to USD 44bn (IAB/PwC via Voxbooster, 2026); 60% of marketers use AI in influencer work
Community and organic social	H, B	Low to medium	Limited: authenticity the top social AI challenge (43%)
Retail and commerce media	D	High	Moderate: about USD 200bn in 2026 (WARC via Manifest Media, 2026); India USD 1.93bn in 2025 (Osmos, 2026)
Connected TV and streaming	B	Medium	Moderate: 63% of marketers see it as a top trend
Email, SMS and push	D	High	Moderate: email leads AI use at 36% (Ascend2); a widely quoted USD 36 return per USD 1 was not traced to a primary source
Messaging and conversational commerce (including WhatsApp)	D, H	High	Moderate: Meta paid messaging above a USD 2bn run rate (PPC Land, 2026)
CRM, loyalty and lifecycle marketing	D, H	High	Limited: 44% of Indian consumers more loyal to trusted brands (Adobe via Storyboard18, n.d.)
Personalization and customer experience	D	High	Moderate: 84% use AI for real-time personalization (Salesforce, as summarized)
Pricing, offers and promotions	D	Medium	Limited: retail budgets reportedly shifting toward margin management (Martech.zone, 2026)
Partnerships, co-marketing, affiliates and referral	H, V	Low to medium	None retrieved
Public relations, earned media, events and sponsorships	H, B	Low to medium	None retrieved
Account-based marketing and intent data (B2B)	D	High	Limited: trade headline only
Product-led growth, reviews and user content	H	Medium	None retrieved
Local and vernacular marketing	B, H	Medium	None retrieved; Adobe reports 96% of Indian consumers expect product videos (Impact, n.d.)

Tactic	Main capital	Role of AI	Evidence in this study
Sustainability and responsible-brand claims	H, B	Low	Limited: trade report of a 12% rise in green marketing budgets
Customer research and insight	D	High	Moderate for direction; synthetic respondent validity unsettled (Peres et al., 2023)
Marketing mix modelling and experimentation	C	High	Moderate: only 41% can prove AI ROI (Konabayev, 2026)

Appendix B. Evidence Base Used in the Audit

B1. Platform and advertiser disclosures

Measure	Value	Date and source
Meta end-to-end AI ad tools annual run rate	About USD 20 billion (up 70% year on year); above USD 60 billion	Q4 2024; Q3 2025 (MediaCat, 2025; Marketing Dive, 2025)
Advertisers using at least one Meta generative AI tool	Over 4 million, up from 1 million in Q2 2024	Q4 2024 (MediaCat, 2025)
Meta video generation tools run rate	USD 10 billion	Q4 2025 (PPC Land, 2026)
Advantage+ sales campaigns, average ROAS improvement	About 22%	2025 (PPC Land, 2026)
Meta average price per ad	Up 10% (Q3 2025); up 6% (Q4 2025)	Marketing Dive, 2025; PPC Land, 2026
Klarna AI savings	About USD 10 million annualized; 37% of marketing savings	May 2024 (Marketing Dive, 2024)

B2. Search studies

Study	Finding
Pew Research Center, July 2025	Click on a traditional result 8% with an AI Overview vs 15% without (Beancount, 2026)
Ahrefs, Dec 2025 data	About 58% lower click-through for the top organic result with an AI Overview (Ahrefs, 2026)
Seer Interactive	Organic click-through on AI Overview queries fell from 1.76% to 0.61%; paid clicks fell 68% (Beancount, 2026)
Similarweb	Zero-click searches rose from 56% to 69%, May 2024 to May 2025 (Beancount, 2026)
DataRefs	AI Overviews on 48% of Google queries in April 2026 (DataRefs, 2026)

B3. Controlled productivity evidence

Study	Result
Noy & Zhang (2023)	ChatGPT cut writing time by 40% and raised rated quality by 18% in 453 professionals
Brynjolfsson, Li & Raymond (2023)	Customer support productivity up 14% on average and 34% for novices (5,179 agents)
Dell'Acqua et al. (2023)	758 consultants: inside the frontier, 12.2% more tasks, 25.1% faster and over 40% higher quality; outside it, 19 percentage points less likely to be correct

Appendix C. Source Reliability Tiers

Tier	Definition	Examples	Use in this paper
A	Peer-reviewed or controlled experiments	Noy & Zhang (2023); Luo et al. (2019); Lambrecht & Tucker (2019)	Causal claims
B	Large independent surveys with disclosed samples	McKinsey Global Survey on AI; Gartner CMO Spend Survey; Pew Research Center	Adoption and behavior trends
C	Vendor, platform and agency reports; company statements	Meta earnings statements, Salesforce, HubSpot, Kantar, Klarna	Directional; sponsor bias
D	Aggregators and republishers	Statistics compilations and trade summaries	Only when no better source; marked "as reported"

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Web sources were retrieved between 2 and 6 October 2026. Statistics from primary publishers (Ascend2, Attest, Capterra, Qualtrics, YouGov, Cloudflare Radar, IAB and others) are cited through the republisher listed, and the primary reports were not read directly.

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